as you’d never use a sorting algorithm with exponential time complexity, it makes sense to think about the cost of scaling.

You should have two expectations when scaling an application:

- The ability to handle any demand increase with finite resources is unrealistic. So, ideally, you want the resources to increase slowly when demand grows: a linear relationship or better. Figure 1.1 shows the relationship between demand and required resources.
- If resources must be increased, ideally, you want the complexity of the application to stay the same or increase slowly. (Remember the good old free lunch when no added complexity was required for a faster application!) Figure 1.2 shows the relationship between resources and complexity.